

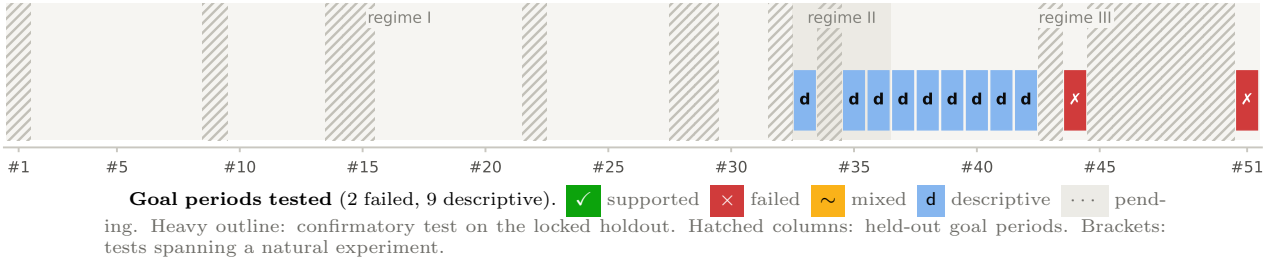
H66: Shared platform latency is a common-noise field

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 45% Faithfulness 2/5 Usefulness 1.5/5

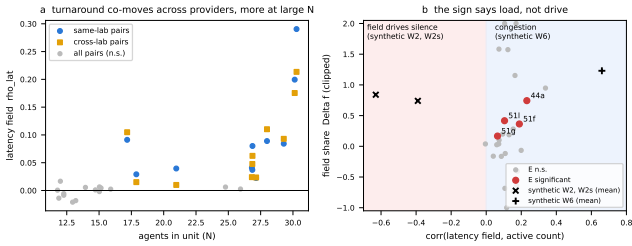
Bottom line. The residual co-activation after trimming is small; the platform’s common latency field is real but follows activity rather than driving it.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

After the day is trimmed to the window in which every agent runs, little co-activation is left in regime III. The question is whether a common platform-latency field explains that residual. The test uses call turnaround from the DQ1 ledger on 29 non-holdout units (regimes II and III) and Gemini’s logged server time. The residual beats the trimmed block-shift null in 4/29 units. A latency field exists in 11/29 units, all in #44 and #51, and it grows with N . It is not a provider field: same-lab and cross-lab pairs co-move alike. It is not a driver either. It rises when more agents are active (28/29 units), which is the opposite of the model’s sign. The pre-registered rule says mixed; a post hoc sign test and a congestion world say the field share measures load. **Refuted as posed.** The holdout is not run.



Observables. (a) Cross-agent turnaround correlation per regime-III unit vs N . Same-lab and cross-lab pairs co-move alike, so the field is not a provider’s API. (b) Field share Δf against the sign of the field’s relation to the active count. A field that silences agents gives a negative sign (synthetic W2). Congestion gives a positive sign and $\Delta f \approx 1$ (W6). Every real unit with a residual sits on the load side.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Do not blame platform slowness for agents acting together. On the call clock, the platform’s latency field is a load meter: it rises when more agents run, and it does not make them co-act. A field share from a third-party-field regression is uninformative without the sign of the field–activity relation; report both. For a scaffold builder, the useful number is the growth of turnaround co-movement with N : about 0 at $N \leq 16$, 0.02–0.10 at N 17–28, and 0.2 at $N = 30$. Watch it as a capacity signal. Gemini’s API time is only 0.44 of a call’s turnaround, and it does not co-move across agents; the shared part is the village’s own execution layer.

PHYSICAL MODEL

Models 01 and 02. The spin $a_i(m)$ is 1 when agent i logs an act call in minute m of the all-present window. The latency spin $z_i(m)$ is the agent-day-standardized log turnaround of its chained calls. The field L_{-ij} is the mean of z_k over third parties $k \neq i, j$, with lags 0–5 min, plus an infrastructure-error count u . $h_i(b)$ is a 30-min block field. E is the mean pair correlation of block-demeaned spins above the block-shift null. K is the number of active agents.

$$P(\mathbf{a}) \propto \exp \left[\sum_i (h_i(b) + \sum_{k=0}^5 \gamma_{ik} L(m-k) + \kappa_i u(m)) a_i + \frac{J_0}{2N} (\sum_i a_i)^2 \right]$$
$$f_{\text{lat}} = 1 - E_{\text{adj}}/E, \quad \Delta f = f_{\text{lat}} - \langle f_{\text{lat}} \rangle_{\text{shifted field}}$$

field drives silence: $\text{corr}(L, K) < 0$; load: $\text{corr}(L, K) > 0$

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RESULTS VS PREDICTIONS

Prediction (dated)	Observed	Verdict
P0 synthetic recovery	sizes 0/16, 1/16; Δf a lower bound (W2 0.74 vs 1.0); no false attribution	partly
P1 field in $\geq 2/3$ units; provider field	11/26 units; same-lab 0.080 vs cross-lab 0.062	failed
P2 $\Delta f < 0.25$ where E sig.	4/29 sig.; Δf 0.74, 0.36, 0.17, load (A2) 0.42; rule: mixed	failed
Post hoc sign test	$\text{corr}(L, K) > 0$ in 28/29 (median +0.11); W2 gives -0.34 to -0.66	against H66
P3 zero-lag field peak	$C_\ell(0)$ 0.023 vs $C_\ell(\pm 1)$ 0.017	held
P4 intensity $>$ binary	14/29	failed
P5 same-room $>$ cross-room E	5/18 two-room units	failed
P6 residual at window edges	3/4 units ($\times 2.0$ –6.1)	held
G51 server time (native)	r 0.49 with turnaround; no cross-agent co-movement	failed as H66
G44 separate-stack leader	loading -0.11 vs median 0.14 (70 min)	descriptive

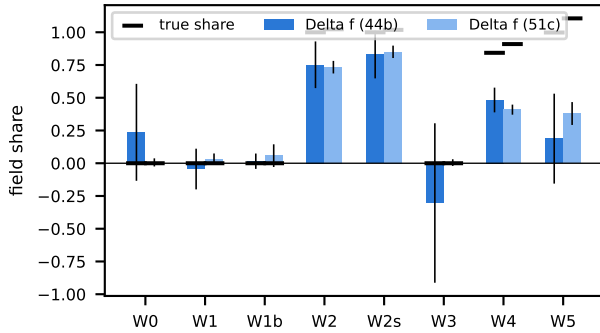
GOAL PERIODS IN DETAIL

G33 d descriptive
G35 d descriptive
G36 d descriptive
G37 d descriptive
G38 d descriptive
G39 d descriptive
G40 d descriptive
G41 d descriptive
G42 d descriptive
G44 $\times$ failed
+1 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	1
E interventional	0		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).



Second figure. Synthetic swarms on real unit skeletons (44b, 51c; 8 seeds). Bars: Δf ; ticks: true field share. The estimator stays near 0 when the field does not act (W1, W3). It recovers 0.4–0.85 of a field that drives pauses (W2, W2s, W4).

CAVEATS

The sign test and the congestion world are post hoc (Amendment 2); they reverse a pre-registered “mixed”. Turnaround mixes generation, tool execution and overhead; only Gemini logs server time. Load may be congestion or shared heavy work; the data do not separate them. Only 4 units carry a residual, and their Δf intervals are wide. The separately served leader has 70 minutes. The holdout is not run.

ROUND 2

Fit a load curve (latency vs active count) per unit and its slope vs N . Use provider-specific error bursts as exogenous latency shocks. Run `confirm.py` after sign-off.

- **Essence.** separating exogenous platform fields from swarm-made fields in the residual co-activation.
- **H66-R1** Congestion as an order parameter.
- **H66-R2** An exogenous platform probe.
- **H66-R3** Separate execution from generation with the Anthropic and OpenAI token counts (output tokens per second) where usage is logged.
- **H66-R4** Holdout.

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